CENG60002: Reaction Engineering II

October 9, 2026

1 Topic 0. Conditions for Phase Stability, and Reaction Equilibrium

Chemical reactors are not at equilibrium, as the net rates of reactions are usually not allowed to become zero. Considering a single-phase multicomponent system, if we declare the following variables: T, temperature, P, pressure, and n_i the number of moles of each component i. The Gibbs energy of the system G utilizes these state variables to declare the conditions for equilibrium,

$$dG = -SdT + VdP + \sum_j \mu_j dn_j$$

where S is the entropy, V is the volume; both of the system, and μ is the chemical potential of component j.

2 Topic 1. External Mass Transfer

The analysis of external mass transfer will be done on a catalyst surface. We define the following variables: $C_{A,s}$, the concentration of reactant A in the catalyst, $C_{A,ss}$: the concentration of reactant A at the surface of the catalyst $z = 0$, and finally C_A: the concentration of A in the bulk liquid.

We can define the following rates:

1. **Mass Transfer**: $r_{A,mass} = k_{m_c}\left(C_A - C_A^{surf}\right)$, where $k_{m_c} = \frac{D_{AB}}{\delta}$ (derived from Fick's law of diffusion), this assumes a linear diffusion profile along the laminar film across the surface of the catalyst.
2. **Reaction Rate**: $r_{A,kinetics} = k_{surf}C_A^{surf}$; simple rate of reaction, where $k_{surf} = k_0 \exp^{\left(-\frac{E_a}{RT}\right)}$

At equilibrium:

$$\begin{aligned} r_{A,\text{mass}} &= r_{A,\text{kinetics}} \\ k_{m_c}\left(C_A - C_A^{\text{surf}}\right) &= k_{\text{surf}}C_A^{\text{surf}} \\ C_A^{surf} &= \left(\frac{k_{m,c}}{k_{m,c} + k_{surf}}\right) C_A \end{aligned}$$

Finally, we define the overall rate constant k_0 as

$$\frac{1}{k_0} = \frac{1}{k_{m_c}} + \frac{1}{k_{surf}}$$

Due to this construction, we can observe the following limiting cases:

1. Rapid Mass Transfer: $k_{m_c} >> k_s$, resulting in $k_0 \approx k_s$, and $C_{A,surf} \approx C_A$, which is therefore rate-controlled.
2. Rapid Reaction: $k_s >> k_{m_c}$, resulting in $k_0 \approx k_{m_c}$, and $C_{A,surf} \approx 0$, which is therefore diffusion-controlled.

There are also several important dimensionless numbers:

1. **Fick's Law Corrections**: $\delta_A = \frac{(r+s+...)-(a+b+...)}{a}$.
2. **Reynolds Number**: $Re = \frac{Gd_p}{\mu} = \frac{inertial\ effects}{viscous\ effects}$.
3. **Prandlt Number**: $Re = \frac{C_p \mu}{\lambda} = \frac{viscous\ effects}{heat\ transfer\ effects}$.
4. **Schmidt Number**: $Sc = \frac{\mu}{\rho_f D} = \frac{viscous\ effects}{diffusion\ effects}$.